Fluid Mechanics

(ME 150)

What is a Fluid?

- *When a constant shear force is applied, a solid eventually* stops deforming, at some fixed strain angle, whereas a fluid never stops deforming and approaches a certain rate of strain.

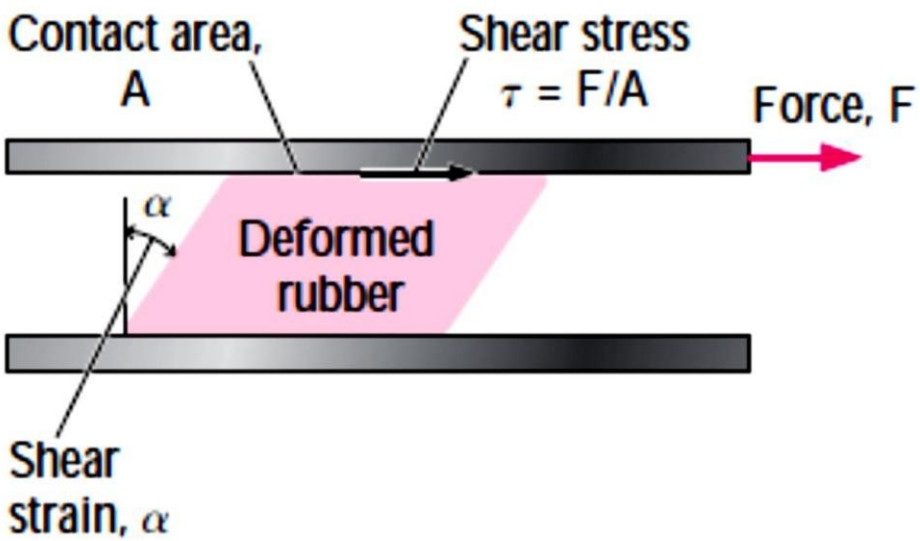
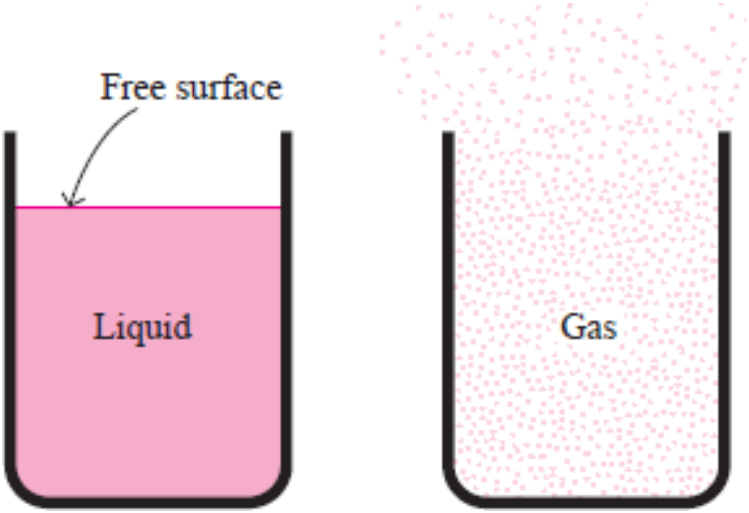


Figure.
Deformation of a rubber eraser placed between two parallel plates under the influence of a shear force.

What is a Fluid?

- In a liquid, molecules can move relative to each other, but the volume remains relatively constant because of the strong cohesive forces between the molecules.
- As a result, a liquid takes the shape of the container it is in, and it forms a free surface in a larger container in a gravitational field.
- A gas, on the other hand, expands until it encounters the walls of the container and fills the entire available space.
- This is because the gas molecules are widely spaced, and the cohesive forces between them are very small.
- Unlike liquids, gases cannot form a free surface



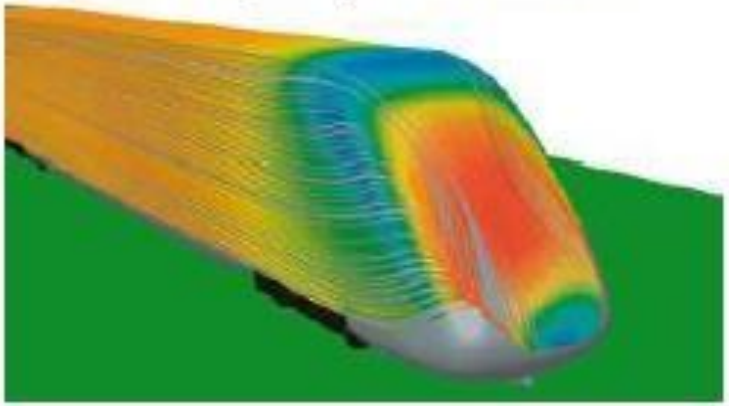
What is a Fluid?

- Differences between liquid and gases

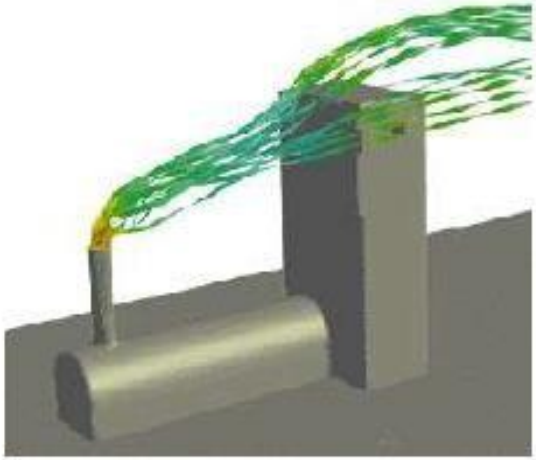
Liquid	Gases
Difficult to compress and often regarded as incompressible	Easily to compress – changes of volume is large, cannot normally be neglected and are related to temperature
Occupies a fixed volume and will take the shape of the container	No fixed volume, it changes volume to expand to fill the containing vessels
A free surface is formed if the volume of container is greater than the liquid.	Completely fill the vessel so that no free surface is formed.

Application areas of Fluid Mechanics

High speed train

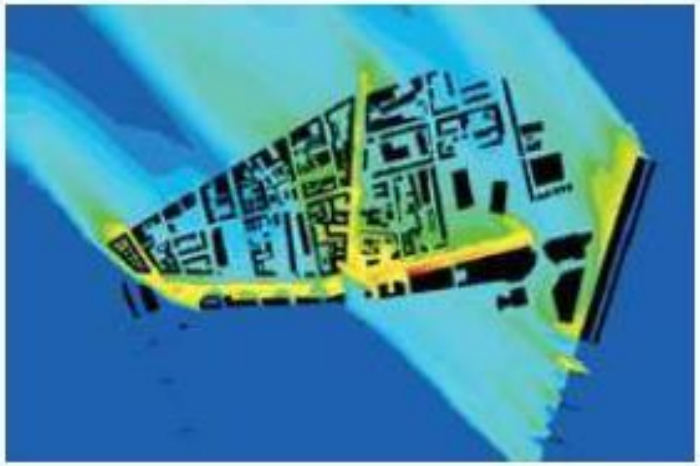


Wind turbines

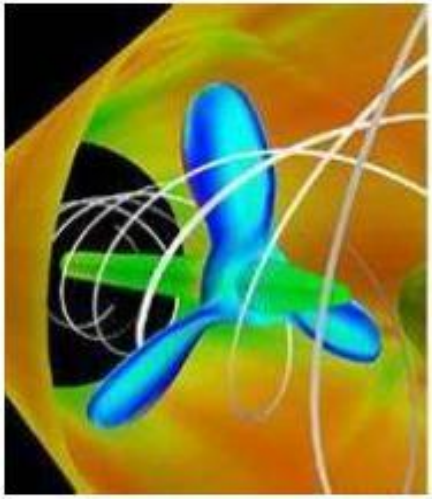


Smoke from a stack

Pollutant dispersion over a city



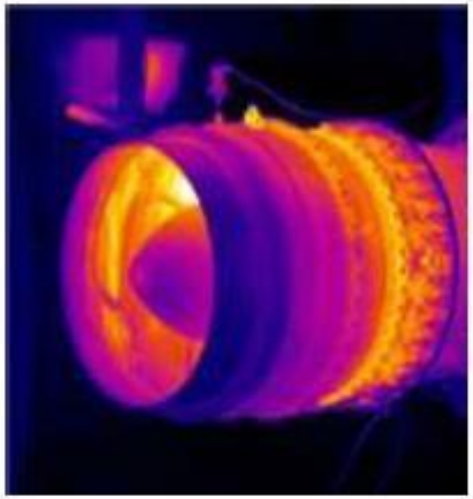
Application areas of Fluid Mechanics



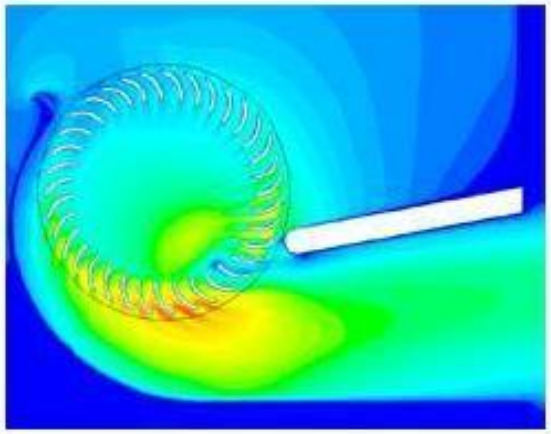
Propeller

Fan

Centrifugal Pump



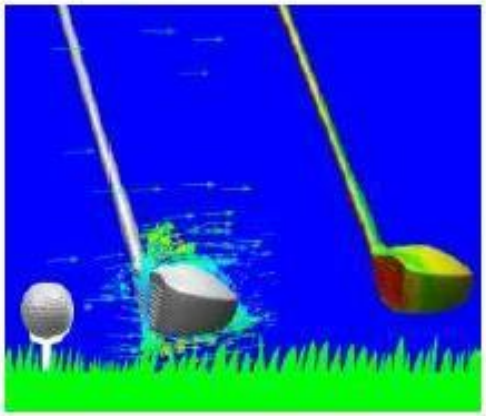
Jet Engine
Propulsion



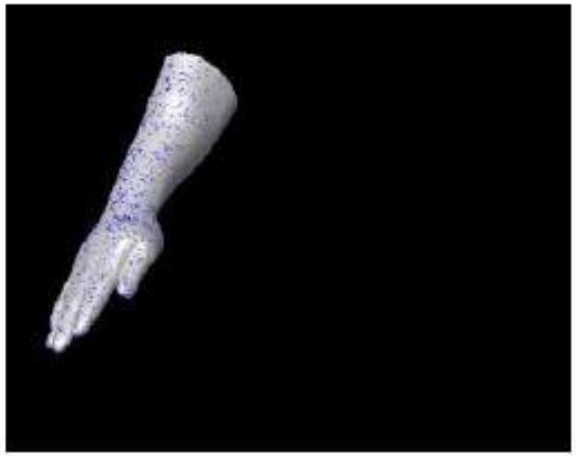
Crossflow fan

Application areas of Fluid Mechanics

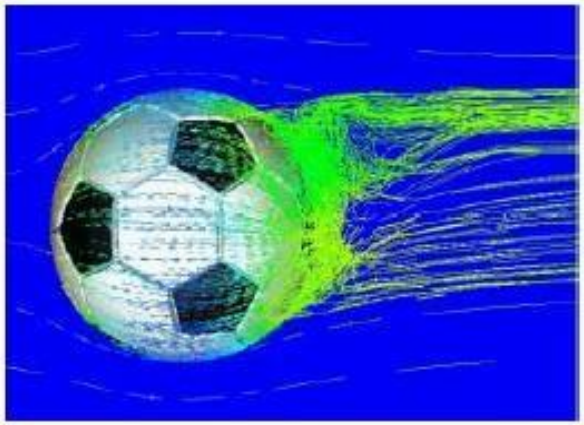
Ski Jumping



Golf

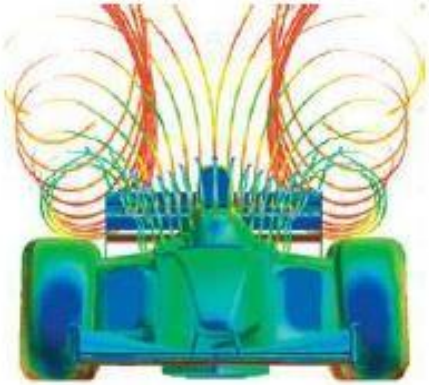


Swimming



Football

Cycling



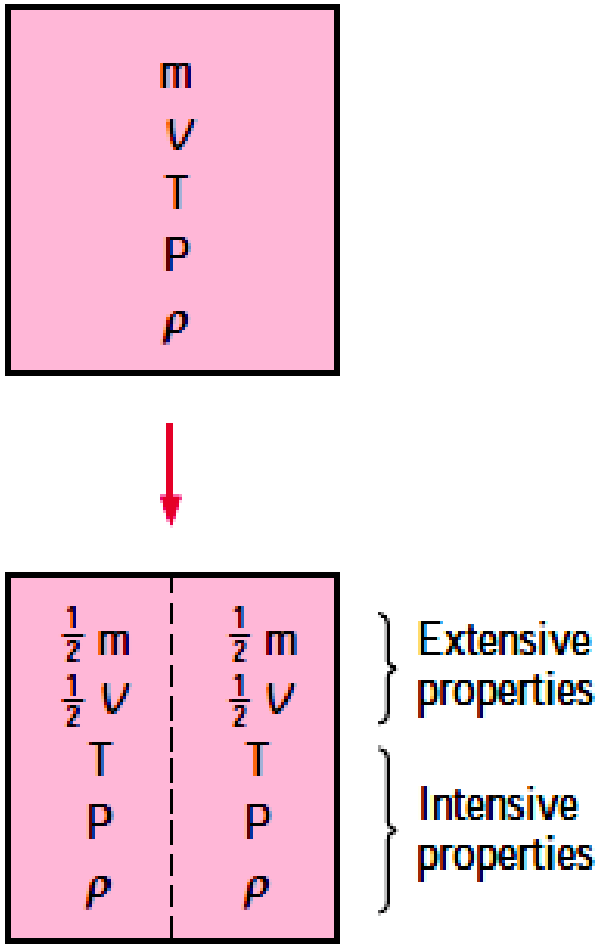
Indy Car Racing

Properties of Fluids

- Any characteristic of a system is called a **property.**
- Some familiar properties are pressure *P*, temperature T, volume V, and mass m.
- Other less familiar properties include viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electric resistivity, and even velocity and elevation.
- Properties are considered to be either ***intensive or extensive***.
- ***Intensive*** properties are those that are independent of the mass of a system, such as temperature, pressure, and density.
- **Extensive** properties are those whose values depend on the size—or extent—of the system. Total mass, total volume *V*, and total momentum are some examples of extensive properties.

Properties of Fluids

- An easy way to determine whether a property is intensive or extensive is to divide the system into two equal parts with an imaginary partition.
- Each part will have the same value of intensive properties as the original system, but half the value of the extensive properties.



Properties of Fluids

Density or Mass Density

- Density or mass density of a fluid is defined as the ratio of the mass of a fluid to its volume. Thus mass per unit volume of a fluid is called density. It is denoted the symbol ρ (rho). The unit of mass density in SI unit is kg per cubic meter, i.e ., kg/m^3.
- The density of liquids may be considered as constant while that of gases changes with the variation of pressure and temperature.
- Mathematically mass density is written as.

$$\rho = \frac{\text{Mass of fluid}}{\text{Volume of fluid}}$$

- The value of density of water is 1 gm/cm^3 or 1000 kg/m^3.

Properties of Fluids

Density or Mass Density

- The density of a substance, in general, depends on temperature and pressure.
- The density of most gases is proportional to pressure and inversely proportional to temperature.
- Liquids and solids, on the other hand, are essentially incompressible substances, and the variation of their density with pressure is usually negligible.

Properties of Fluids

Specific weight or Weight Density

- Specific weight or weight density of a fluid is the ratio between the weight of a fluid to its volume.
- Thus weight per unit volume of a fluid is called weight density and it is denoted by the symbol w.
- Mathematically,

$$w = \frac{\text{Weight of fluid}}{\text{Volume of fluid}} = \frac{\text{(Mass of fluid) x Acceleration due to gravity}}{\text{Volume of fluid}}$$

$$= \frac{\text{Mass of fluid x g}}{\text{Volume of fluid}}$$

$$= \rho \text{ x g}$$

$$w = \rho g$$

Properties of Fluids

Specific Volume

- Specific volume of a fluid is defined as the volume of a fluid occupied by a unit mass or volume per unit mass of a fluid is called specific volume.
- Mathematically, it is expressed as

$$\text{Specific volume} = \frac{\text{Volume of fluid}}{\text{Mass of fluid}} = \frac{1}{\dfrac{\text{Mass of fluid}}{\text{Volume}}} = \frac{1}{\rho}$$

- Thus specific volume is the reciprocal of mass density. It is expressed as m^3/kg.
- It is commonly applied to gases.

Properties of Fluids

Specific Gravity.

- Specific gravity is defined as the ratio of the weight density (or density) of a fluid to the weight density (or density) of a standard fluid.
- For liquids, the standard fluid is taken water and for gases, the standard fluid is taken air. Specific gravity is also called relative density. It is dimensionless quantity and is denoted by the symbol *S*.

$$S(\text{for liquids}) = \frac{\text{Weight density (density) of liquid}}{\text{Weight density (density) of water}}$$

$$S(\text{for gases}) = \frac{\text{Weight density (density) of gas}}{\text{Weight density (density) of air}}$$

$$\text{Thus weight density of a liquid} = S \text{ x Weight density of water}$$

$$= S \text{ x } 1000 \text{ x } 9.81 \text{ N/m}^3$$

$$\text{Thus density of a liquid} = S \text{ x Density of water}$$

$$= S \text{ x } 1000 \text{ kg/m}^3$$

Properties of Fluids

Specific Gravity.

- If the specific gravity of a fluid is known, then the density of the fluid will be equal to specific gravity of fluid multiplied by the density of water.
- For example the specific gravity of mercury is 13.6, hence density of mercury = 13.6 x 1000 = 13600 kg/m^3.

Specific gravities of some substances at 0°C

Substance	SG
Water	1.0
Blood	1.05
Seawater	1.025
Gasoline	0.7
Ethyl alcohol	0.79
Mercury	13.6
Wood	0.3–0.9
Gold	19.2
Bones	1.7–2.0
Ice	0.92
Air (at 1 atm)	0.0013

Properties of Fluids

Example 1.
Calculate the specific weight, density and specific gravity of one liter of a liquid which weighs 7 N.

Solution. Given :

$$\text{Volume} = 1 \text{ litre} = \frac{1}{1000} \text{ m}^3 \quad \left(\because \ 1 \text{ litre} = \frac{1}{1000} \text{ m}^3 \text{ or } 1 \text{ litre} = 1000 \text{ cm}^3\right)$$

$$\text{Weight} = 7 \text{ N}$$

(*i*) Specific weight (w) $= \dfrac{\text{Weight}}{\text{Volume}} = \dfrac{7 \text{ N}}{\left(\dfrac{1}{1000}\right) \text{m}^3} = \mathbf{7000 \text{ N/m}^3}$. **Ans.**

(*ii*) Density (ρ) $= \dfrac{w}{g} = \dfrac{7000}{9.81} \text{ kg/m}^3 = \mathbf{713.5 \text{ kg/m}^3}$. **Ans.**

(*iii*) Specific gravity $= \dfrac{\text{Density of liquid}}{\text{Density of water}} = \dfrac{713.5}{1000}$ $\{\because$ Density of water $= 1000 \text{ kg/m}^3\}$

$= \mathbf{0.7135}$. **Ans.**

Example 2. *Calculate the density, specific weight and weight of one liter of petrol of specific gravity = 0.7*

Solution. Given : Volume = 1 litre = $1 \times 1000\ cm^3 = \frac{1000}{10^6}\ m^3 = 0.001\ m^3$

Sp. gravity $S = 0.7$

(*i*) *Density* (ρ)

Density (ρ) $= S \times 1000\ kg/m^3 = 0.7 \times 1000 =$ **700 kg/m³. Ans.**

(*ii*) *Specific weight* (*w*)

$w = \rho \times g = 700 \times 9.81\ N/m^3 =$ **6867 N/m³. Ans.**

(*iii*) *Weight* (*W*)

We know that specific weight $= \frac{\text{Weight}}{\text{Volume}}$

or $w = \frac{W}{0.001}$ or $6867 = \frac{W}{0.001}$

$\therefore$ $W = 6867 \times 0.001 =$ **6.867 N. Ans.**

Properties of Fluids

Viscosity

- Viscosity is defined as the property of a fluid which offers resistance to the movement of one layer of fluid over another adjacent layer of the fluid.
- When two layers of a fluid, a distance *'dy' apart move one* over the other at different velocities say u and u+ *du as shown in Fig. 1.1 , the viscosity together with* relative velocity causes a shear stress acting between the fluid layers:

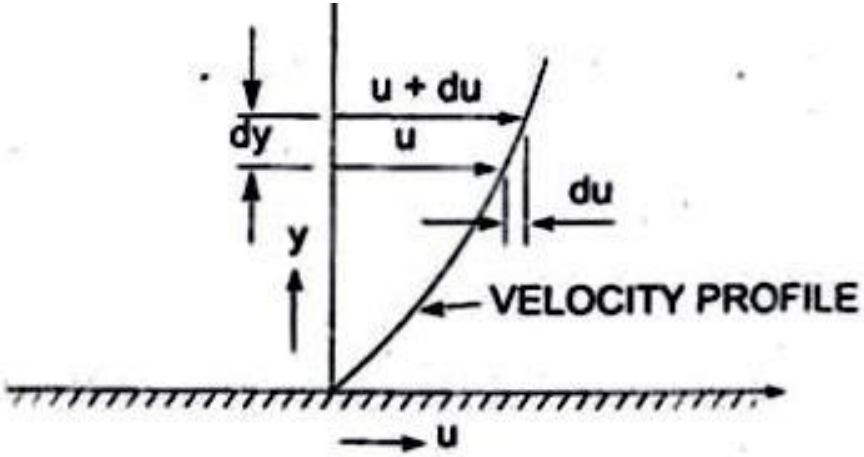


Fig. 1.1 ***Velocity variation near a solid boundary.***

Properties of Fluids

Viscosity

- The top layer causes a shear stress on the adjacent lower layer while the lower layer causes a shear stress on the adjacent top layer.
- This shear stress is proportional to the rate of change of velocity with respect to *y. It is denoted by symbol* τ called Tau.
- Mathematically,

$$\tau \propto \frac{du}{dy}$$

- or

$$\boxed{\tau = \mu \frac{du}{dy} \qquad (1.2)}$$

Properties of Fluids

- where μ (called m*u) is the constant of proportionality and is known as the coefficient of dynamic viscosity* or only viscosity.
- $\frac{du}{dy}$ *represents the rate of shear strain or rate of shear deformation or velocity gradient.*
- *From equation (1.2) we have*

$$\mu = \frac{\tau}{\frac{du}{dy}} \qquad (1.3)$$

- Thus viscosity is also defined as the shear stress required to produce unit rate of shear strain.

Properties of Fluids

Unit of Viscosity.

- The unit of viscosity is obtained by putting the dimension of the quantities in equation (1.3)

$$\mu = \frac{\text{Shear stress}}{\dfrac{\text{Change of velocity}}{\text{Change of distance}}} = \frac{\text{Force/Area}}{\left(\dfrac{\text{Length}}{\text{Time}}\right) \times \dfrac{1}{\text{Length}}}$$

$$= \frac{\text{Force}/(\text{length})^2}{\dfrac{1}{\text{Time}}} = \frac{\text{Force} \times \text{Time}}{(\text{Length})^2}$$

$$\text{SI unit of viscosity} = \frac{\text{Newton second}}{m^2} = \frac{Ns}{m^2}$$

Properties of Fluids

Kinematic Viscosity.

- It is defined as the ratio between the dynamic viscosity and density of fluid.lt is denoted by the Greek symbol (ν) called 'nu' . Thus, mathematically,

$$\nu = \frac{\text{Viscosity}}{\text{Density}} = \frac{\mu}{\rho}$$

- The SI unit of kinematic viscosity is m^2/s.

Newton's Law of Viscosity.

- It states that the shear stress (τ) on a fluid element layer is directly proportional to the rate of shear strain. The constant of proportionality is called the co-efficient viscosity. Mathematically, it is expressed as given by equation (1 . 2).

Properties of Fluids

- Fluids which obey the above relation are known as Newtonian fluids and the fluids which do not obey the above relation are called Non-newtonian fluids.

Variation of Viscosity with Temperature

- Temperature affects the viscosity.
- The viscosity of liquids decreases with the increase of temperature while the viscosity of gases increases with increase of temperature. This is due to reason that the viscous forces in a fluid are due to cohesive forces and *molecular momentum transfer.*
- *In liquids the cohesive forces predominates the molecular momentum* transfer due to closely packed molecules and with the increase in temperature, the cohesive forces decreases with the result of decreasing viscosity.

Properties of Fluids

- But in the case of gases the cohesive force are small and molecular momentum transfer predominates. With the increase in temperature, molecular momentum transfer increases and hence viscosity increases. The relation between viscosity and temperature for liquids and gases are:

(i) For liquids, $\mu = \mu_o \left(\frac{1}{1 + \alpha t + \beta t^2} \right)$

where μ = Viscosity of liquid at t°C, in poise $\qquad 1 \text{ poise} = \frac{1}{10} \frac{Ns}{m^2}$

μ_o = Viscosity of liquid at $0°C$, in poise

α, β = are constants for the liquid

For water, $\mu_o = 1.79 \times 10_{-3}$ poise, $\alpha = 0.03368$ and $\beta = 0.000221$

(ii) For a gas, $\mu = \mu_o + \alpha t - \beta t_2$

where for air $\mu_o = 0.000017$, $\alpha = 0.000000056$, $\beta = 0.1189 \times 10_{-9}$

Types of Fluids

1. Ideal Fluid. A fluid, which is incompressible and is having no viscosity, is known as an ideal fluid. Ideal fluid is only an imaginary fluid as all the fluids, which exist, have some viscosity.
2. Real fluid. A fluid, which possesses viscosity, is known as real fluid. All the fluids: in actual practice, are real fluids.
3. Newtonian Fluid. A real fluid, in which the shear stress is directly, proportional to the rate of shear strain (or velocity gradient), is known as a Newtonian fluid.
4. Non-Newtonian fluid. A real fluid, in which shear stress is not proportional to the rate of shear strain (or velocity gradient), known as a Non-Newtonian fluid.

Types of Fluids

5. Ideal Plastic Fluid.

   A fluid, in which shear stress is more than the yield value and shear stress is proportional to the rate of shear strain (or velocity gradient), is known as ideal plastic fluid.

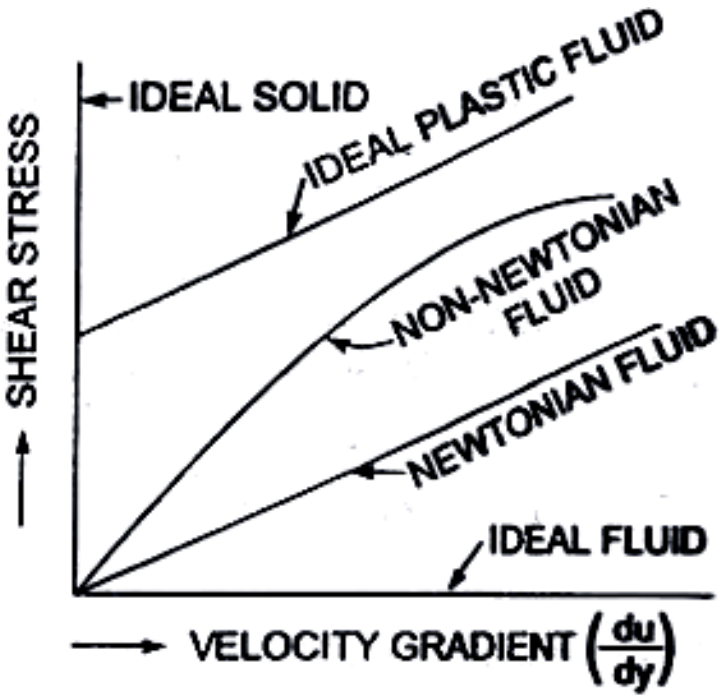


Fig. 1.2 *Types of fluids.*

Example

If the velocity distribution over a plate is given by

$$u = \frac{2}{3}y - y^2$$

in which u is velocity in metre per second at a distance y metre above the plate, determine the shear stress at y = 0 and y= 0.15 m. Take dynamic viscosity of fluid as 8.63 poises.

Solution. Given : $u = \frac{2}{3}y - y^2 \quad \therefore \quad \frac{du}{dy} = \frac{2}{3} - 2y$

$$\left(\frac{du}{dy}\right)_{\text{at } y=0} \text{ or } \left(\frac{du}{dy}\right)_{y=0} = \frac{2}{3} - 2(0) = \frac{2}{3} = 0.667$$

Also $$\left(\frac{du}{dy}\right)_{\text{at } y=0.15} \text{ or } \left(\frac{du}{dy}\right)_{y=0.15} = \frac{2}{3} - 2 \times .15 = .667 - .30 = 0.367$$

Value of $\mu = 8.63$ poise $= \frac{8.63}{10}$ SI units $= 0.863$ N s/m^2

Now shear stress is given by equation (1.2) as $\tau = \mu \frac{du}{dy}$.

(*i*) Shear stress at $y = 0$ is given by

$$\tau_0 = \mu \left(\frac{du}{dy}\right)_{y=0} = 0.863 \times 0.667 = \mathbf{0.5756\ N/m^2.\ Ans.}$$

(*ii*) Shear stress at $y = 0.15$ m is given by

$$(\tau)_{y=0.15} = \mu \left(\frac{du}{dy}\right)_{y=0.15} = 0.863 \times 0.367 = \mathbf{0.3167\ N/m^2.\ Ans.}$$

Example

Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size 0.8 m x 0.8 m and an inclined plane with angle of inclination 30° as shown in Fig. 1.4. The weight of the square plate is 300 N and it slides down the inclined plane with a uniform velocity of 0.3 m/s. The thickness of oil film is 1.5 mm.

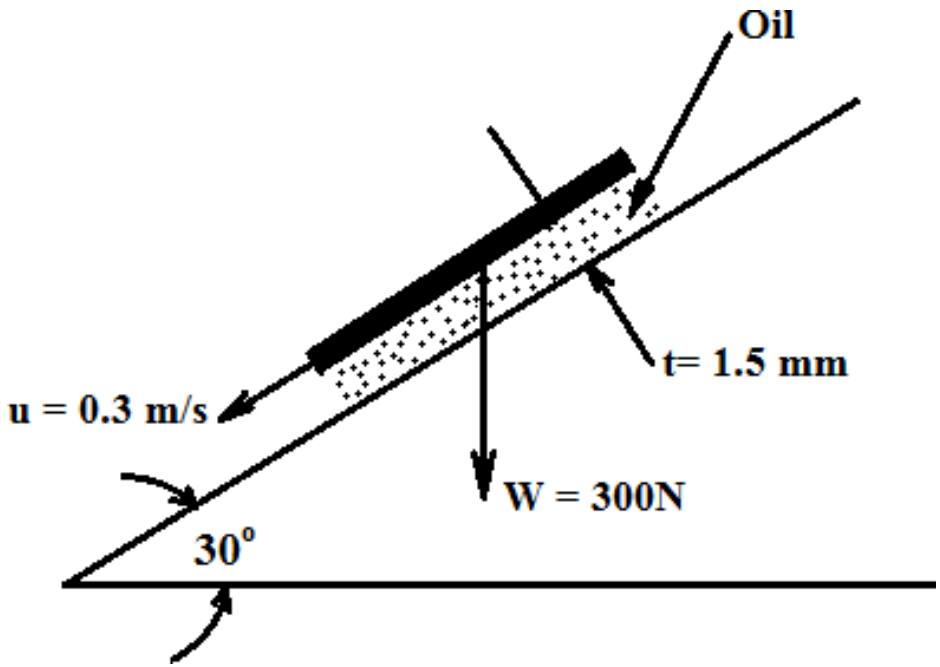


Fig.1.4

Solution. Given :

Area of plate, $A = 0.8 \times 0.8 = 0.64 \text{ m}^2$

Angle of plane, $\theta = 30°$

Weight of plate, $W = 300 \text{ N}$

Velocity of plate, $u = 0.3 \text{ m/s}$

Thickness of oil film, $t = dy = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$

Let the viscosity of fluid between plate and inclined plane is μ.

Component of weight W, along the plane $= W \cos 60° = 300 \cos 60° = 150 \text{ N}$

Thus the shear force, F, on the bottom surface of the plate = 150 N

and shear stress,
$$\tau = \frac{F}{\text{Area}} = \frac{150}{0.64} \text{ N/m}^2$$

Now using equation (1.2), we have

$$\tau = \mu \frac{du}{dy}$$

where du = change of velocity $= u - 0 = u = 0.3$ m/s

$$dy = t = 1.5 \times 10^{-3} \text{ m}$$

$$\therefore \quad \frac{150}{0.64} = \mu \frac{0.3}{1.5 \times 10^{-3}}$$

$$\therefore \quad \mu = \frac{150 \times 1.5 \times 10^{-3}}{0.64 \times 0.3} = 1.17 \text{ N s/m}^2 = 1.17 \times 10 = \mathbf{11.7 \text{ poise. Ans.}}$$

Example

The space between two square flat parallel plates is filled with oil. Each side of the plate is 60 cm. The thickness of the oil film is 12.5 mm. The upper plate, which moves at 2.5 metre per sec requires a force of 98.1 N to maintain the speed.
Determine : ·
i.the dynamic viscosity of the oil, and
ii.the kinematic viscosity of the oil if the specific gravity of the oil is 0.95.

Solution. Given:
Each side of a square plate = 60 cm = 0.6 m
Area A= 0.6 x 0.6 = 0.36 m^2
Thickness of oil film dy = 12.5 mm = 12.5 x 10^{-3} m
Velocity of upper plate u = 2.5 m/s

$\therefore$ Change of velocity between plates, $du = 2.5$ m/sec

Force required on upper plate, $F = 98.1$ N

$\therefore$ Shear stress, $$\tau = \frac{\text{Force}}{\text{Area}} = \frac{F}{A} = \frac{98.1\text{ N}}{0.36\text{ m}^2}$$

(*i*) Let μ = Dynamic viscosity of oil

Using equation (1.2), $$\tau = \mu \frac{du}{dy} \text{ or } \frac{98.1}{0.36} = \mu \times \frac{2.5}{12.5 \times 10^{-3}}$$

$\therefore$ $$\mu = \frac{98.1}{0.36} \times \frac{12.5 \times 10^{-3}}{2.5} = 1.3635 \frac{\text{Ns}}{\text{m}^2} \quad \textbf{Ans.}$$

(*ii*) Sp. gr. of oil, S = 0.95

Let ν = kinematic viscosity of oil

Using equation (1.1 A),

Mass density of oil, $$\rho = S \times 1000 = 0.95 \times 1000 = 950 \text{ kg/m}^3$$

Using the relation, $\nu = \frac{\mu}{\rho}$, we get $$\nu = \frac{1.3635\left(\frac{\text{Ns}}{\text{m}^2}\right)}{950} = .001435 \text{ m}^2/\text{sec} \quad \textbf{Ans.}$$

Thermodynamic Properties

- Fluids consist of liquids or gases. But gases are compressible fluids and hence thermodynamic properties play an important role.
- With the change of pressure and temperature, the gases undergo large variation in density.
- The relationship between pressure (absolute), specific volume and temperature (absolute) of a gas is given by the equation of state as

$$\boxed{p\forall = RT \text{ or } \frac{p}{\rho} = RT}$$

where p = Absolute pressure of a gas in N/m^2

$\forall$ = Specific volume = $\frac{1}{\rho}$

R = Gas constant

T = Absolute temperature in °K

ρ = Density of a gas.

Thermodynamic Properties

- The value of gas constant R is $R = 287 \frac{J}{kg.K}$

- **Isothermal Process**. If the changes in density occurs at constant temperature, then the process is called isothermal and relationship between pressure (p) and density (ρ) is given by

$$\frac{p}{\rho} = \text{constant}$$

- **Adiabatic Process.** If the change in density occurs with no heat exchange to and from the gas, the process is called adiabatic. And if no heat is generated within the gas due to friction, the relationship between pressure and density is given by

$$\frac{p}{\rho^k} = \text{constant}$$

Thermodynamic Properties

- where k = *Ratio of specific heat of a gas at constant pressure and constant volume.*
- k = 1.4 for air

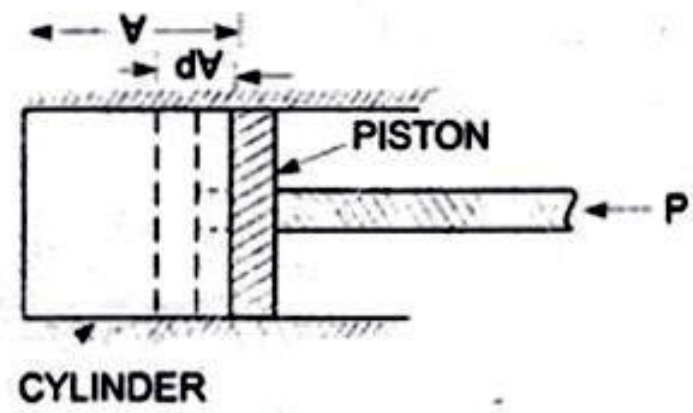
Compressibility and Bulk Modulus

- Compressibility is the reciprocal of the bulk modulus of elasticity, K which is defined as the ratio of compressive stress to volumetric strain.
- Consider a cylinder fitted with a piston as shown in the Fig.
- Let V= *Volume of a gas enclosed in the cylinder*

  p =*Pressure of gas when volume is* V
- Let the pressure is increased *to* $p+dp$, *the volume of gas* decreases from V *to* $V-dV$.
- Then increase in pressure = dp
- Decrease in volume = dV
- Volumetric strain = - dV/V

Compressibility and Bulk Modulus

- - ve sign means the volume decreases with increase of pressure.



$$\therefore \text{Bulk modules} \quad K = \frac{\text{Increase of pressure}}{\text{Volumetric strain}}$$

$$= \frac{dp}{\frac{-dV}{V}} = -\frac{dp}{dV}V$$

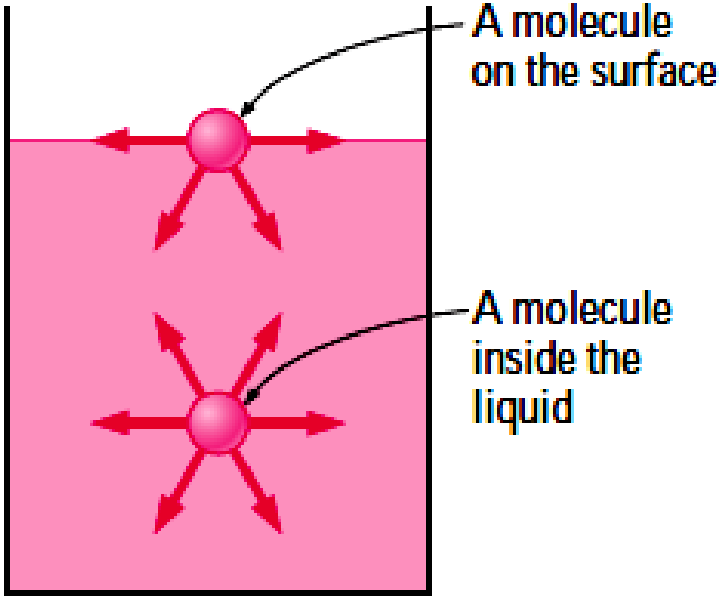
- Compressibility is given by = 1/K

Surface Tension and Capillarity

- Surface tension is defined as the tensile force acting on the surface of a liquid in contact with a gas or on the surface between two immiscible liquids such that the contact surface behaves like a membrane under tension.
- Surface tension is created due to the unbalanced cohesive forces acting on the liquid molecules at the fluid surface.
- Molecules in the interior of the fluid mass are surrounded by molecules that are attracted to each other equally.
- However, molecules along the surface are subjected to a net force toward the interior.
- The apparent physical consequence of this unbalanced force along the surface is to create the hypothetical skin or membrane.

Surface Tension and Capillarity

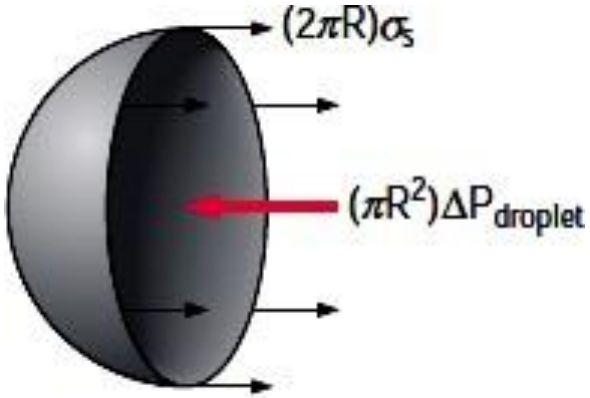
- A tensile force may be considered to be acting in the plane of the surface along any line in the surface.
- The intensity of the molecular attraction per unit length along any line in the surface is called the ***surface tension.***
- It is denoted by Greek letter σ (called sigma).
- The SI unit is N/m.



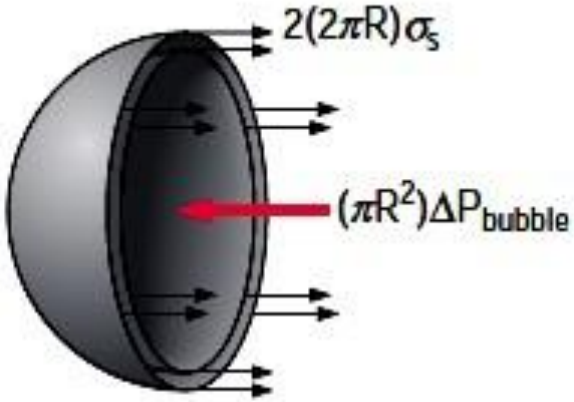
Surface Tension and Capillarity

Surface Tension on liquid Droplet and Bubble

- Consider a small spherical droplet of a liquid of radius *'R'. On the entire surface of the droplet, the tensile force due to surface tension will be acting.*
- Let σ = surface tension of the liquid
- ΔP= Pressure intensity inside the droplet (in excess of the outside pressure intensity)
- *R= Radius of droplet.*
- Let the droplet is cut into two halves. The forces acting on one half (say left half) will be



(a) Half a droplet



(b) Half a bubble

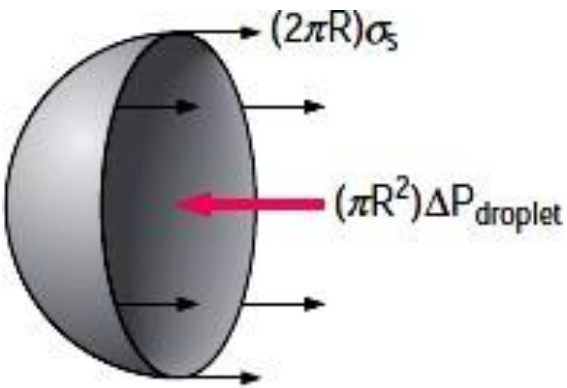
Surface Tension and Capillarity

- (i) tensile force due to surface tension acting around the circumference of the cut portion as shown and this is equal to

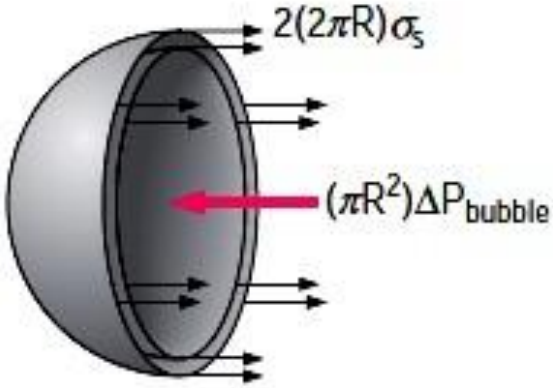
  $= \sigma \times \text{Circumference}$

  $= \sigma \times 2\pi R$

- (ii) pressure force on the area $(\pi/4)d^2$ and
- $= \Delta P \; x \; \pi R^2$ *as shown*



(a) Half a droplet



(b) Half a bubble

Surface Tension and Capillarity

- These two forces will be equal and opposite under equilibrium conditions, *i.e.*,

Droplet: $$(2\pi R)\sigma_s = (\pi R^2)\Delta P_{droplet} \rightarrow \Delta P_{droplet} = P_i - P_o = \frac{2\sigma_s}{R}$$

Bubble: $$2(2\pi R)\sigma_s = (\pi R^2)\Delta P_{bubble} \rightarrow \Delta P_{bubble} = P_i - P_o = \frac{4\sigma_s}{R}$$

- A hollow bubble like a soap bubble in air has two surfaces in contact with air, one inside and other outside. Thus two surfaces are subjected surface tension.

Surface Tension……. Example 1

- Find the surface tension in a soap bubble of 40 mm diameter when the inside pressure is 2.5 N/m^2 above atmospheric pressure.

Solution. Given :

Dia. of bubble, $d = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$

Pressure in excess of outside, $p = 2.5 \text{ N/m}^2$

For a soap bubble, using equation (1.15), we get

$$p = \frac{8\sigma}{d} \quad \text{or} \quad 2.5 = \frac{8 \times \sigma}{40 \times 10^{-3}}$$

$$\sigma = \frac{2.5 \times 40 \times 10^{-3}}{8} \text{ N/m} = \mathbf{0.0125 \text{ N/m. Ans.}}$$

Surface Tension……. Example

- The pressure outside the droplet of water of diameter 0.04 mm is 10.32 N/cm^2 (atmospheric pressure). Calculate the pressure within the droplet if surface tension is given as 0.0725 N/m of water.

Surface Tension and Capillarity

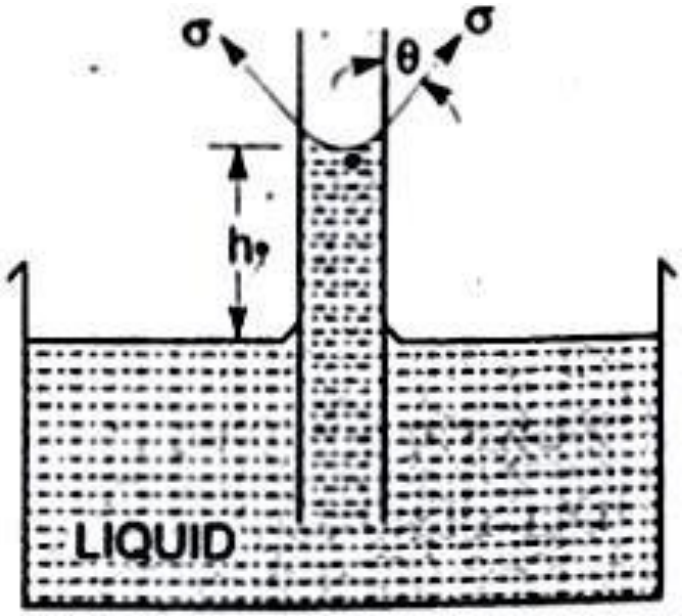
Capillarity

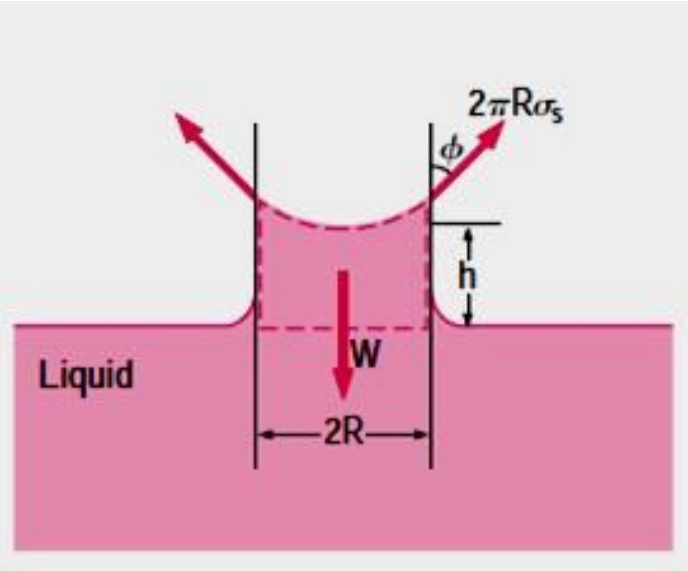
- Capillarity is defined as a phenomenon of rise or fall of a liquid surface in a small tube relative to the adjacent general level of liquid when the tube is held vertically in the liquid.
- The rise of liquid surface is known as capillary rise while the fall of the liquid surface is known as capillary depression.
- The attraction (adhesion) between the wall of the tube and liquid molecules is strong enough to overcome the mutual attraction (cohesion) of the molecules and pull them up the wall. Hence, the liquid is said to *wet the solid surface.*
- It is expressed in terms of cm or mm of liquid. Its value depends upon the specific weight of the liquid, diameter of the tube and surface tension of the liquid.

Surface Tension and Capillarity

Expression for Capillary Rise

- Consider a glass tube of small diameter ‘d’ opened at both ends and is inserted in a liquid, say water.
- The liquid will rise in the tube above the level of the liquid.
- Let h = the height of the liquid in the tube . Under a state of equilibrium, the weight of the liquid of height h is balanced by the force at the surface of the liquid in the tube. But the force at the surface of the liquid in the tube is due to surface tension.





Expression for Capillary Rise…

- Let σ = Surface tension of liquid

  θ = Angle of contact between the liquid and glass tube

- The weight of the liquid of height h in the tube

  = (Area of the tube x h) x ρ x g

$$= \frac{\pi}{4} d^2 \times h \times \rho \times g$$

where ρ = Density of liquid

Vertical component of the surface tensile force

$$= (\sigma \times \text{Circumference}) \times \cos\theta$$

$$= \sigma \times \pi d \times \cos\theta$$

For equilibrium, equating (1.17) and (1.18), we get

$$\frac{\pi}{4} d^2 \times h \times \rho \times g = \sigma \times \pi d \times \cos\theta$$

or

$$h = \frac{\sigma \times \pi d \times \cos\theta}{\frac{\pi}{4} d^2 \times \rho \times g} = \frac{4\sigma\cos\theta}{\rho \times g \times d}$$

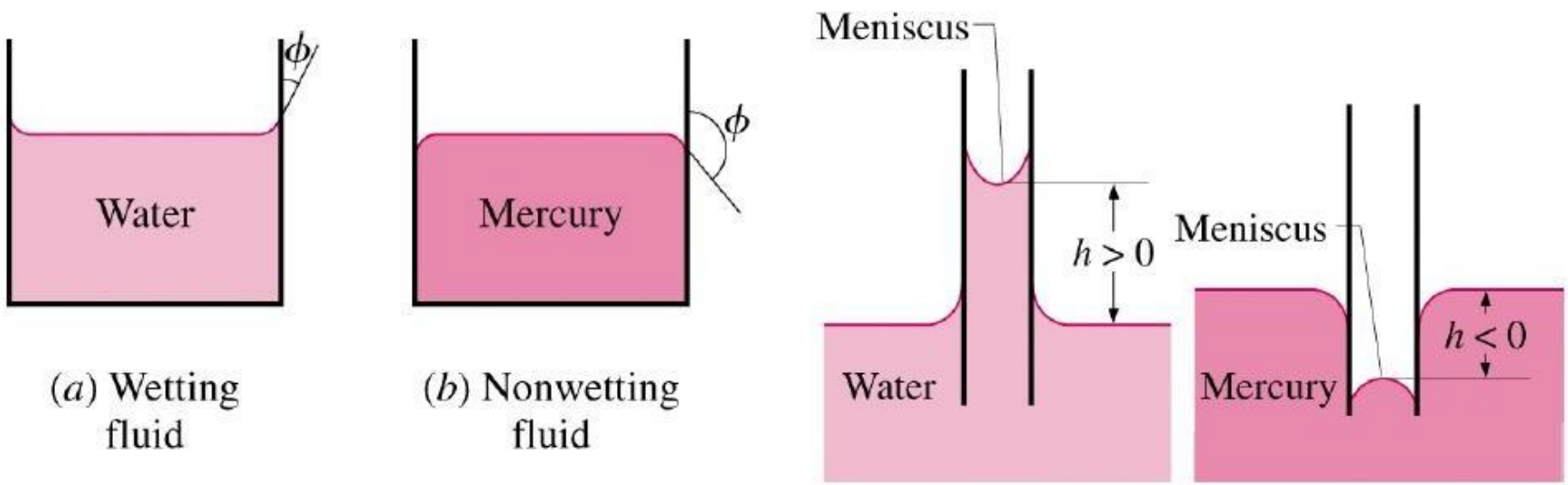
Expression for Capillary Rise…

- The value of θ between water and clean glass tube is approximately equal to zero and hence cos θ is equal to unity. Then rise of water is given by

$$h = \frac{4\sigma}{\rho \times g \times d}$$

- Contact angle depends on both the liquid and the solid.
- If θ is less than 90°, the liquid is said to "wet" the solid. However, if θ is greater than 90°, the liquid is repelled by the solid, and tries not to "wet" it.
- For example, water wets glass, but not wax. Mercury on the other hand does not wet glass.

Capillarity



(*a*) Wetting fluid

(*b*) Nonwetting fluid

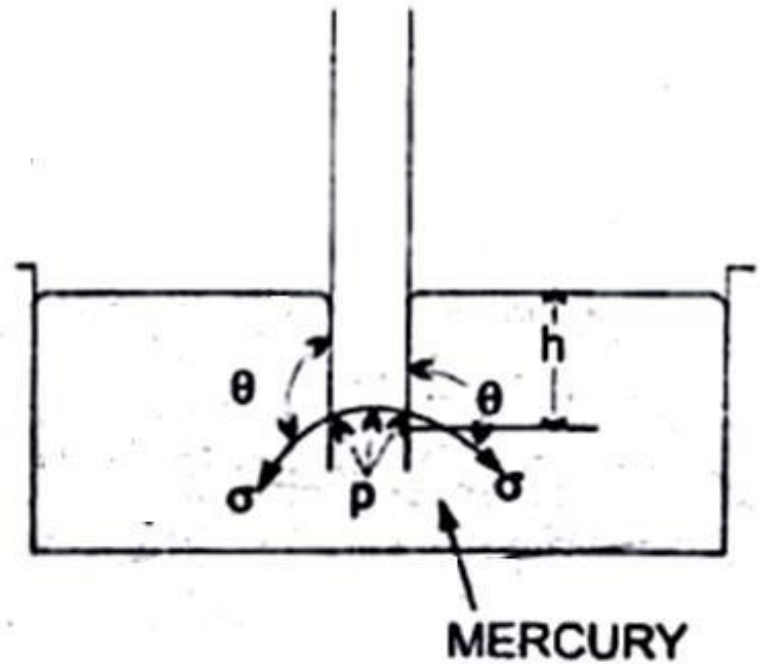
Expression for Capillary Fall

- If the glass tube is dipped in mercury, the revel of mercury in the tube will be lower than the general level of the outside liquid as shown above.

Capillarity

Expression for Capillary Fall

- Let h = *Height of depression in tube.*
- Then in equilibrium, two forces arc acting on the mercury inside the tube.
- First one is due to surface tension acting in the downward direction and is equal to $\sigma \times \pi d \times \cos\theta$.
- Second force is due to hydrostatic force acting upward and is equal to intensity of pressure at a depth *'h' x Area*



Capillarity

Expression for Capillary Fall

$$= p \times \frac{\pi}{4} d^2 = \rho g \times h \times \frac{\pi}{4} d^2 \; \{\because p = \rho g h\}$$

Equating the two, we get

$$\sigma \times \pi d \times \cos\theta = \rho g h \times \frac{\pi}{4} d^2$$

$$h = \frac{4\sigma \cos\theta}{\rho g d}$$

Value of θ for mercury and glass tube is 128°

Capillarity…Example 1

- Calculate the capillary rise in a glass tube of 2.5 mm diameter when immersed vertically in (a) water and (b) mercury. Take surface tensions $\sigma = 0.0725$ N/m for water and $\sigma = 0.52$ N/m for mercury in contact with air. The specific gravity for mercury is given as 13.6 and angle of contact = 130°.

Solution. Given :

Dia. of tube,	$d = 2.5 \text{ mm} = 2.5 \times 10^{-3} \text{ m}$
Surface tenstion, σ for water	$= 0.0725$ N/m
σ for mercury	$= 0.52$ N/m
Sp. gr. of mercury	$= 13.6$

Capillarity…Example

$\therefore$ Density $= 13.6 \times 1000 \text{ kg/m}^3$.

(*a*) Capillary rise for water ($\theta = 0$)

Using equation (1.20), we get $h = \dfrac{4\sigma}{\rho \times g \times d} = \dfrac{4 \times 0.0725}{1000 \times 9.81 \times 2.5 \times 10^{-3}}$

$= .0118 \text{ m} =$ **1.18 cm. Ans.**

(*b*) For mercury

Angle of constant between mercury and glass tube, $\theta = 130°$

Using equation (1.21), we get $h = \dfrac{4\sigma \cos\theta}{\rho \times g \times d} = \dfrac{4 \times 0.52 \times \cos 130°}{13.6 \times 1000 \times 9.81 \times 2.5 \times 10^{-3}}$

$= -.004 \text{ m} =$ **– 0.4 cm. Ans.**

The negative sign indicates the capillary depression.

Capillarity…Example

- Find out the minimum size of glass tube that can be used to measure water level if the capillary rise in the tube is to be restricted to 2 mm. Consider surface tension of water in contact with air as 0.073575 N/m.

Solution. Given :

Capillary rise, $h = 2.0 \text{ mm} = 2.0 \times 10^{-3} \text{ m}$

Surface tension, $\sigma = 0.073575 \text{ N/m}$

Let dia. of tube $= d$

The angle θ for water $= 0$

The density for water, $\rho = 1000 \text{ kg/m}^3$

$$h = \frac{4\sigma}{\rho \times g \times d} \text{ or } 2.0 \times 10^{-3} = \frac{4 \times 0.073575}{1000 \times 9.81 \times d}$$

$$\therefore \quad d = \frac{4 \times 0.073575}{1000 \times 9.81 \times 2 \times 10^{-3}} = 0.015 \text{ m} = \mathbf{1.5 \text{ cm. Ans.}}$$

Thus minimum diameter of the tube should be 1.5 cm.

Dimensions and Units

- Fluid mechanics deals with the measurement of many variables of many different types of units. Hence we need to be very careful to be consistent.

Dimensions and Base Units

- The *dimension of a measure is independent of any particular system of units. For* example, velocity may be inmetres per second or miles per hour, but dimensionally, it is always length per time, or $L/T = LT^{-1}$.
- The dimensions of the relevant base units of the Système International (SI) system are:

Dimensions and Units

Unit-Free		SI Units	
Dimension	**Symbol**	**Unit**	**Symbol**
Mass	M	kilogram	kg
Length	L	metre	m
Time	T	second	s
Temperature	θ	kelvin	K

Derived Units

Quantity	Dimension	SI Unit	
		Derived	**Base**
Velocity	LT^{-1}	m/s	$m\,s^{-1}$
Acceleration	LT^{-2}	m/s^2	$m\,s^{-2}$
Force	MLT^{-2}	Newton, N	$kg\,m\,s^{-2}$

Pressure Stress	$ML^{-1}T^{2}$	Pascal, Pa N/m^2	$kg\ m^{-1}\ s^{-2}$
Density	ML^{-3}	kg/m^3	$kg\ m^{-3}$
Specific weight	$ML^{-2}T^{-2}$	N/m^3	$kg\ m^{-2}\ s^{-2}$
Relative density	Ratio	Ratio	Ratio
Viscosity	$ML^{-1}T^{-1}$	Ns/m^2	$kg\ m^{-1}\ s^{-1}$
Energy (work)	$ML^{2}T^{-2}$	Joule, J Nm	$kg\ m^{2}\ s^{-2}$
Power	$ML^{2}T^{-3}$	Watt, W Nm/s	$kg\ m^{2}\ s^{-3}$

Note: The acceleration due to gravity will always be taken as 9.81 m/s^2.

Unit Table

Quantity	SI Unit	English Unit
Length (L)	Meter (*m*)	Foot (*ft*)
Mass (m)	Kilogram (*kg*)	Slug (slug) = $lb*sec^2/ft$
Time (T)	Second (*s*)	Second (*sec*)
Temperature (θ)	Celcius (^oC)	Farenheit (^oF)
Force	Newton $(N)=kg*m/s^2$	Pound (lb)

Dimensions and Units…

- 1 *Newton* – Force required to accelerate a 1 *kg* of mass to 1 m/s^2
- 1 *slug* – is the mass that accelerates at 1 ft/s^2 when acted upon by a force of 1 *lb*
- To remember units of a Newton use F=ma (Newton's 2nd Law)
  - [F] = [m][a]= $kg*m/s^2 = N$
- To remember units of a slug also use F=ma => m = F / a
- [m] = [F] / [a] = $lb / (ft / sec^2) = lb*sec^2 / ft$